

Looking for a semantic core in LLM representations (Marco Broni)

Motivation

As a human, 2 very different sentences convey the same MEANING

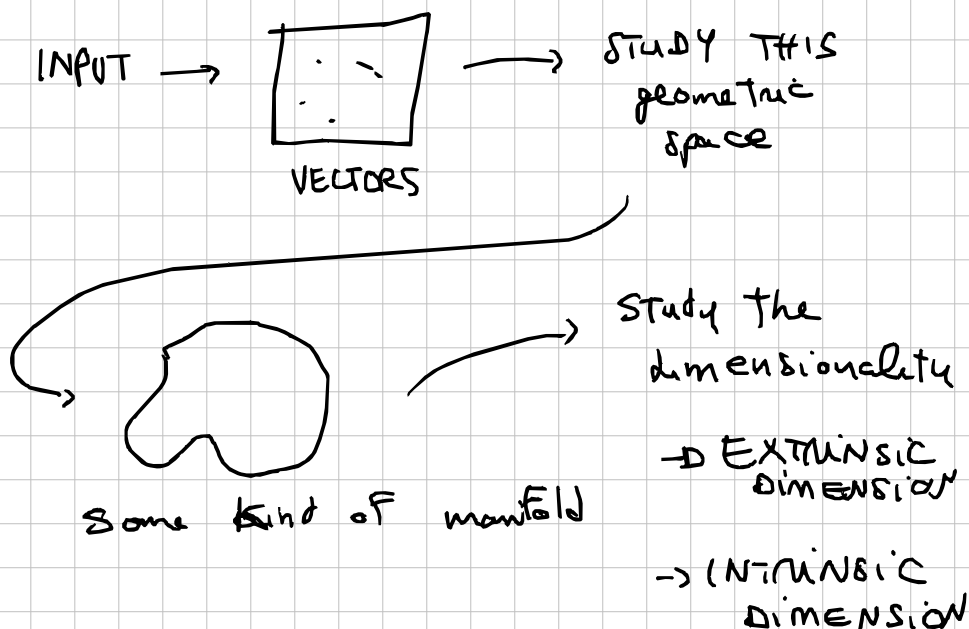
↓
Core to HUMAN LANGUAGE
but LATENT

The question is, can we teach different NN to capture the MEANING of language

↓
Platonic Representation Hypothesis

- 1) Is There really a shared core?
- 2) What are the properties of the semantic core across networks, network layers, modalities and input

Emergence of a High Dimensional



Evolving ID across layers $\rightarrow$ ROBES

layer N $\rightarrow$ layer $N+1$

Also those manifolds are only locally linear, but generally in non-linearity

$\downarrow$

NOT assume linearity

GRIDE estimator (Denti, 2022)

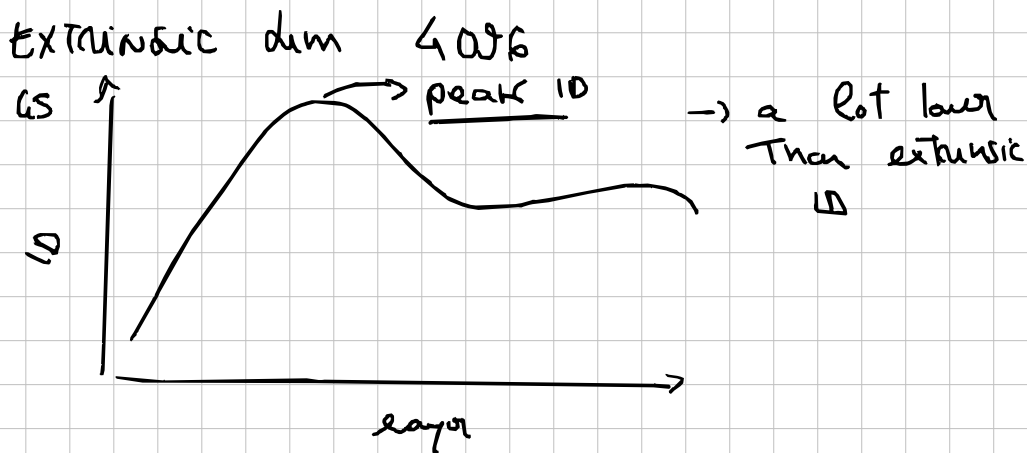
$\rightarrow$ # parameter needed to characterize the data

DADAPY

Random sample of sentences

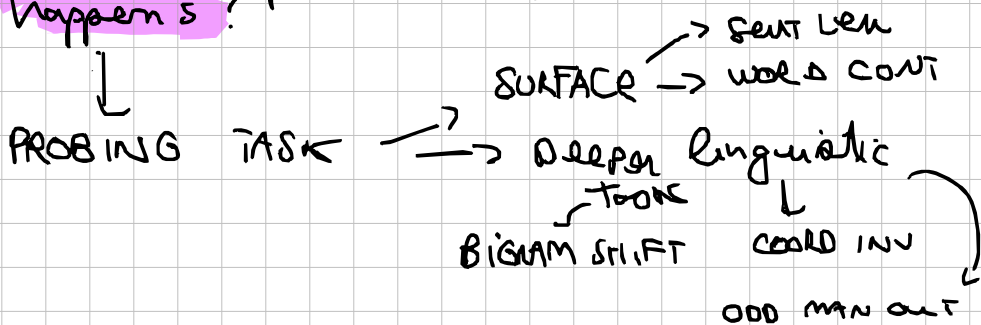
- 10K Fixed-Length tokens from the P.1e
- Also shuffled corpus

→ LAST TOKEN as a representation of a sentence → why CAUSAL LLM



ID peak emerges during Training
→ in various pythia checkpoints

Does ID peak happen when something happens?



→ No correlation btw ID peak and SURFACE PROPERTIES

→ Correlation with deeper probes

MAIN TAKEAWAYS

SYSTEMATIC STUDY OF ID

gzip
Linear
method
→ Compression
paper

→ Extracting representation
↳ with hooks → TOO complex

A QUANTITATIVE ANALYSIS OF SEMANTIC INFORMATION IN DEEP REPRESENTATIONS

↓
VISUAL - TEXTUAL DATA

→ Different
extrinsic
run

→ Random
seed initializa-
tion

→ SIMILARITY STRUCTURES

Neighborhood overlap, Information
Imbalance, Representation Similarity Analysis

Information Imbalance

→ Local Measure, The Austrialek Representation Hypothesis

→ Asymmetric Measure

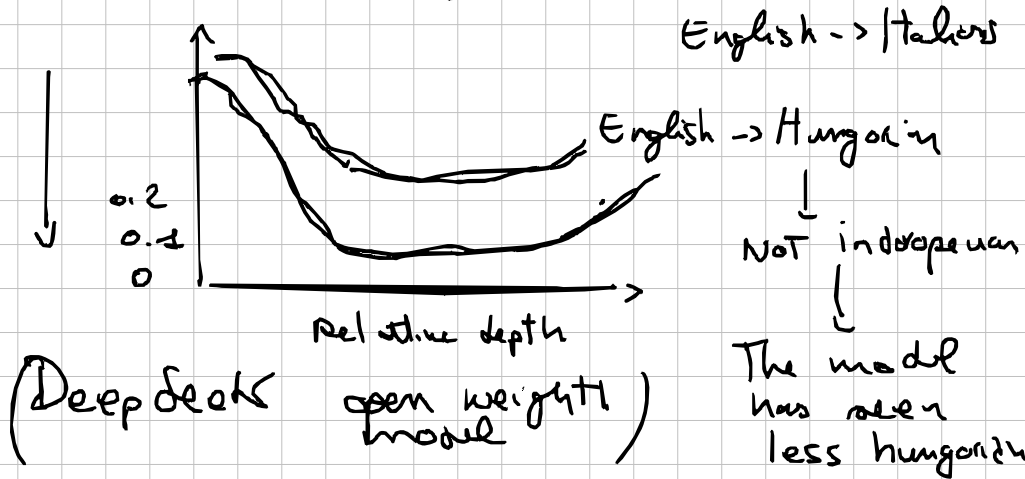
Formulas
$$\Delta(A \rightarrow B) = \frac{2}{N} \sum_{i,j} \frac{r_{ij} \cdot \delta}{v_{ij}^A}$$

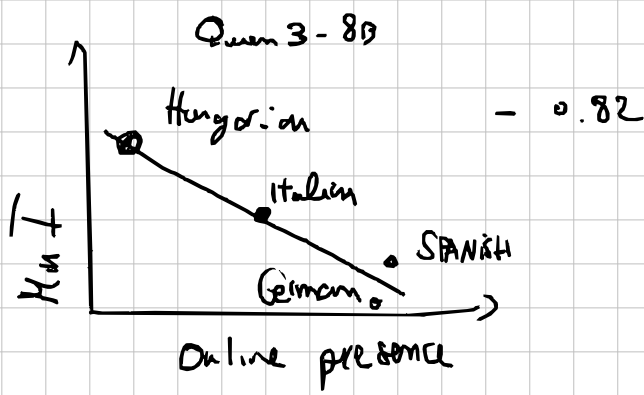
0 → 1

0 → same space

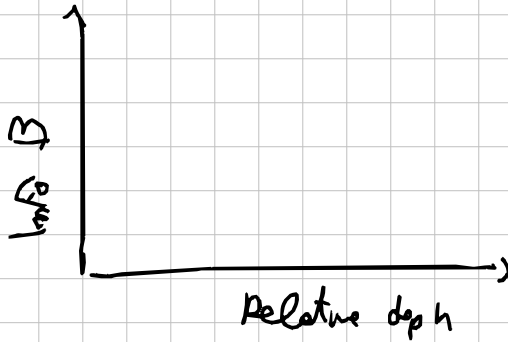
→ Interesting properties (Del Taito 2024)
Dab-pr

Semantic convergence of Translations





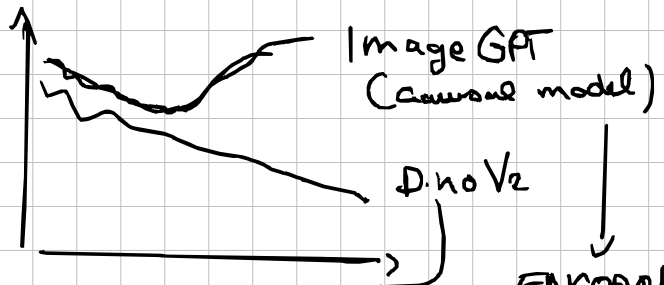
Convergence across models



Can we do the same for visual models?

OBJECTS, images

ImageGPT
DinoV2



To Find representation that are good for many tasks → FEATURE EXTRACTOR

Tied weights and untied weights

→ The model seems to learn

Images and their descriptions in unrelated
visual and language models

CLIP-Text → CLIP-Image

Language predicting vision
is better than the opposite

Summary

Differential syntactics and semantic
encoding in UMs

Syntax twins

↓

Syntax centroids

Semantic twins

↓

Semantic
centroids

Semantic ablation

Syntax ablation

Symmetrized INFORMATION
IMBALANCE

remaining
semantics

>

remaining
syntax

PROBING → SYNTAX
→ semantic retrieval

SYNTAX
semantic
Residual

→ NOT capturing
all

↓
semantic core

Main Takeaways

- Semantic core in central layers or layers
- Ablating syntax seems to disrupt semantics

Now we want to be able to
map to a SHARED
SEMANTIC manifold

↓
composed meaning

How one simple concept retrieved
here → POLAR PROBES

